

LFP2Vec across labs: a reproduction, a preprocessing fix, and a calibration check

Keshav Rajput, September 2026. Code, data manifests, exact splits and every number below: <https://github.com/happyc0der/lfp2vec-audit>. Corrected 21 September 2026; see the README.

Question

LFP2Vec (He et al., NeurIPS 2025) fine-tunes an audio model on raw local field potential to predict which brain region an electrode sits in, and reports zero-shot transfer between labs. Its Broader Impact section states that clinical use “should include calibrated uncertainty estimates”. This note reproduces the fine-tuning stage on the two public datasets the paper uses, asks why cross-lab transfer failed in the reproduction, fixes it, and measures calibration under lab shift, which the paper does not report.

Setup

Data. Seven IBL insertions from five labs and ten Allen Visual Coding probes, chunked as the released code does: three seconds, one hundred chunks per channel. 132 600 IBL and 50 200 Allen chunks. Two Allen probe files that contain only zeros were excluded; IBL contains no CA2 channels, so results use four classes. **Method.** facebook/wav2vec2-base fine-tuned with the paper’s hyper-parameters. The self-supervised continuation is not run; in the released code it runs per dataset and never sees two labs. No pretrained weights are available, so published numbers are read from figures (± 0.01). **Evaluation.** Leave-one-session-out within lab; across labs, train on one and score every probe of the other. Balanced accuracy with chance computed from the classes each test set contains, and raw accuracy over the majority-class rate where the paper uses that. Permuted-label controls on every fold sit at chance.

Results

1. Within a lab it reproduces. 0.74 ± 0.07 balanced accuracy over all seven held-out IBL sessions against the paper’s 0.68; with the paper’s post-processing, 0.81. Fine-tuning adds 0.04 over the untrained audio checkpoint with a linear head (5 of 7 sessions, $p = 0.22$), and replacing inputs with surrogates that keep the power spectrum and destroy the waveform costs 0.008: on this task the model reads spectral power. It beats six-band power by 0.30 on every session ($p = 0.016$).

2. Across labs the full-band model collapses, confidently. It predicts one class for 94% of the other lab’s chunks at 0.98 confidence, landing below the majority rate in both directions on every one of three seeds. A linear probe recovers the source lab from its embeddings at AUC 0.997 on unseen probes. The source is measurable: the datasets’ spectra agree below 100 Hz and diverge up to 768-fold above 300 Hz, because one pipeline band-passes and the other does not.

3. Matching the filters restores transfer. Low-passing every input at 100 Hz, a corner fixed before any cross-lab result existed, with no labels from the target lab:

raw accuracy minus majority-class rate	IBL $\rightarrow$ Allen	Allen $\rightarrow$ IBL
LFP2Vec as published (<i>Figure 2e</i>)	+0.11	+0.12
fine-tuned, full band (3 seeds)	-0.01 ± 0.01	-0.04 ± 0.04
fine-tuned, filters matched, + paper’s post-processing (3 seeds)	+0.08 $\pm$ 0.07	+0.13 $\pm$ 0.03
untrained checkpoint, filters matched, linear head, + post-processing	+0.09	+0.13
band power + paper’s post-processing	+0.17	+0.03
electrode position	-0.24	0.00

Balanced accuracy goes from 0.24 and 0.27 to 0.44 ± 0.10 and 0.37 ± 0.03 , the best of any method tested; lab identity falls from 0.998 to 0.82 and 0.78. The margin matches the published value from Allen to IBL on every seed and falls short of it on average from IBL to Allen (seeds +0.15, +0.02, +0.08). The untrained checkpoint on the same harmonised inputs, with nothing but a linear head fitted, reaches the same margins as the fine-tuned mean with no training of the encoder. Ruled out along the way: the paper’s post-processing alone (0.004 on a collapsed model), per-probe embedding centering (removes the lab signature, leaves transfer at chance), and whitening each probe’s whole spectrum (worse than the low-pass in both directions).

4. Calibration does not survive the shift. Expected calibration error is 0.10 in lab and 0.56 across labs. A temperature fitted in lab cuts the former to 0.03 and moves the latter to 0.51; the target lab needed a temperature four to seven times larger, which requires the labels zero-shot transfer avoids. Matching the filters halves the cross-lab error (0.35) and makes distance from the training distribution a working abstention score in both directions.

5. Signal and position are complementary. Depth along the shank alone reaches 0.77 on IBL, where insertions are stereotyped, 0.49 on Allen, and chance across labs. Fused with the signal as a product of experts it is at or above both parts. When insertion depth is uncertain, position falls ($0.75 \rightarrow 0.63$ at $\pm 400 \mu\text{m}$ on IBL) while the signal stays at 0.68: the signal is the better single source beyond about $\pm 280 \mu\text{m}$.

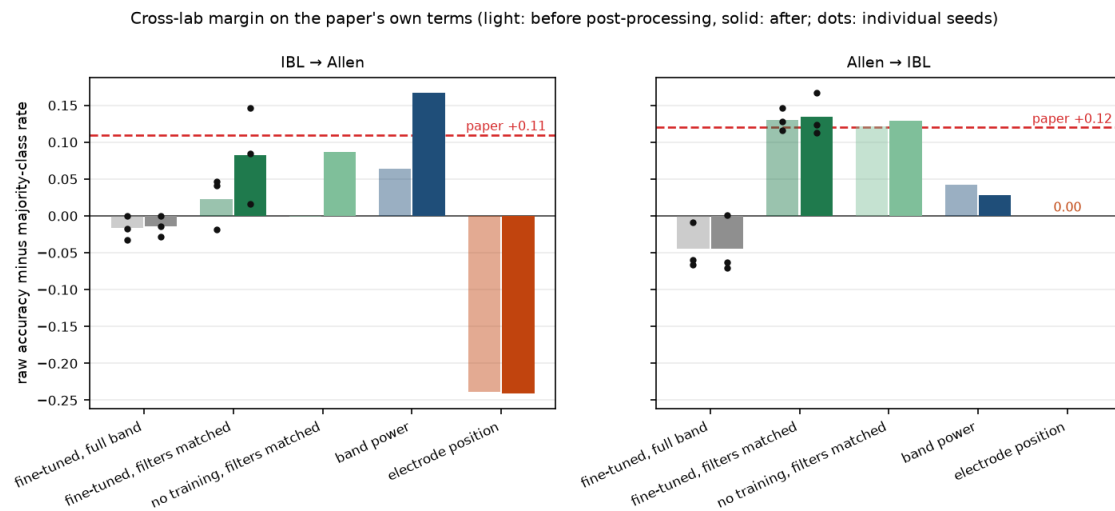


Figure 1: Cross-lab margin on the paper's own metric. Light bars before the paper's post-processing, solid after; dashed line is the published margin.

What this says, and one next experiment

The reduced method does what the paper reports within a lab. Its cross-lab failure here was preprocessing the two datasets do not share, not the representation, and one line of filtering recovers the published margin in one direction and most of it in the other, with or without training the encoder. Confidence is the part that does not transfer, and the measurement the paper calls for is the one that shows it. **Next.** Run the paper's full pipeline, self-supervision included, on inputs harmonised first, and report calibration error beside accuracy. If lab identity stays near 0.8 the residual is physiology; if it falls toward 0.5, acquisition was the whole story.

Limitations

Two of the paper's four datasets are private and untested. Labels are histological estimates taken as given. Within-lab fine-tunes have one seed per session and were not run on Allen; cross-lab fine-tunes have three seeds. Published numbers are read from bitmaps. An earlier version of this note reported a position baseline that leaked test labels; it is corrected here. Much of the code was generated with Claude; every experiment, split and number was independently checked.